ArmorGalv®

FLORIDA CORROSION CHALLENGE

10,000 Hrs

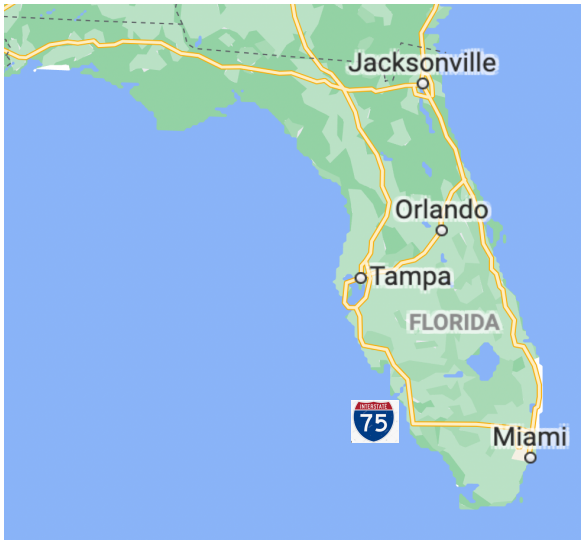
Document #F03

The Florida Corrosion Challenge – 10,000Hrs NSS

FLORIDA STATISTICS

Total coastline: 8,436 miles (13,500Km)
Land mass length: 447 miles (720Km)
Land mass width: 80 miles (130Km)
Average humidity: 74.5%
Average temperature: 76F/24.5C

Most of Florida's population and infrastructure are close to the coastlines. The whole state has high exposure to marine environment.



Salt in the air

The aerosol coming from the sea during monsoon months was found to contain 74% NaCl. The salt concentration was found to increase exponentially with wind speed. The coefficients of the exponential fit are compared with measurements made elsewhere. The gulf cost of Florida is the most corrosive environment in the US, and therefore, the US Navy does much of its corrosion testing in Panama City Florida, where ArmorGalv protected hardware for its fleet of hovercraft was tested and approved.

It is obvious that corrosion of infrastructure is a serious problem for the State of Florida.

The Florida Department of Transportation (**FDOT**) has traditionally specified the use of Hot Dip Galvanize to combat corrosion.

In 2013, to find a more corrosion resistant coating, the **FDOT** undertook to compare **HDG** with **ArmorGalv® 3000**. The result of their testing was published in:
FDOT - CORROSION RESEARCH LABORATORY REPORT 352-955-6691 Jan 10, 2014

The following are some excerpts from this report. The full report is available for download at:
http://armorgalv.com/uploads/downloads/45.pdf

Evaluation of ArmorGalv ® under Salt Fog Exposure and Partial Immersion in Saltwater

Background

ArmorGalv® is a coating achieved through a thermal diffusion galvanizing process similar to sherardizing that reportedly provides corrosion protection, wear resistance, and anti- galling properties. It is applied in accordance with ASTM A1059. The vaporized zinc penetrates the surface of the steel to become integrated with the substrate material. The process provides a consistent thickness which can be applied between 0.4 to 6 mils. The

ArmorGalv®coating is hard, non-magnetic, and can be welded. In addition, the treated product can be bent without causing damage to the coating.

Methodology

Thirteen pieces of steel reinforcement conforming to ASTM A615 were sent to Chem-Plate

Industries to receive the ArmorGalv® process and thirteen pieces were sent to be hot dipped galvanized. Steel reinforcement was used because it is readily available, low cost steel and not because FDOT has plans to use this coating in reinforcement. Three test specimens of each type were subjected to two different exposure conditions and visually evaluated periodically. The first condition was partial immersion in 3.5% saltwater at room temperature (approximately 73F) and the second was in a 5% salt fog at 95F. After 1,500 and 3,000 hours of salt fog exposure a specimen of each type was cross sectioned and mounted for metallographic analysis.

Results

The salt fog exposure revealed large differences between the two specimen types. The hot dipped galvanized bars exhibit almost complete Zinc consumption on all three specimens. (Figure3.)
The ArmorGalv® process showed limited visual signs of corrosion. (Figure 4.) There are several rust-colored stains that are most likely the result of surface contamination.

Figure 3 Traditional Hot Dipped galvanizing after 10,700 hours in salt fog exposure

Figure 4 Armorgalv® process after 10,700 hours in salt fog exposure

Conclusions

*The **ArmorGalv®** coating inhibited corrosion when exposed to salt fog and partial immersion environments very well. It outperformed control samples that were coated utilizing the traditional hot dip galvanizing process.*

NOTES:

- The **ArmorGalv®** coating was subsequently approved for use on bridges in the Jacksonville area, the first one being the San Pablo River bridge.

Excerpt From Print:

5. The new steel posts, steel railing sleeves, anchor bolts, nuts and washers shall be galvanized in accordance with ASTM A1059.

Bridge No. 720366

- Following the FDOT test, the national Academy of science, NCHRP, with support of 7 DOT's, funded and extensive test program of ArmorGalv coated rebar. The test report can be viewed at: http://armorgalv.com/uploads/downloads/37.pdf

  The report makes strong recommendations for using ArmorGalv protected rebar in concrete infrastructure, as replacement for stainless rebar and in high exposure areas along the coasts. It also recommends the use of ArmorGalv protected high tensile rebar.